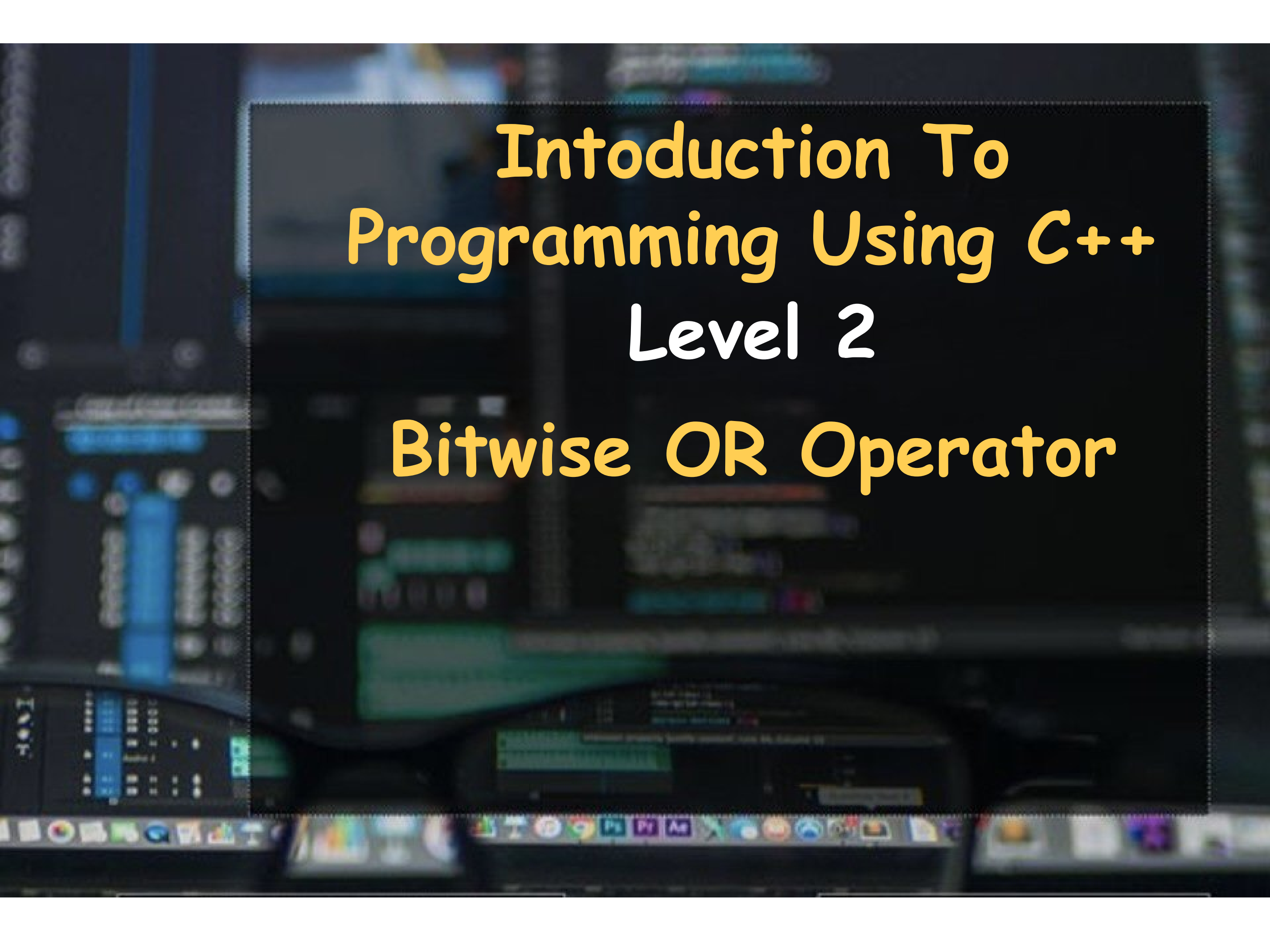


The background of the slide is a blurred image of a computer monitor. On the left side, a calculator application is visible, showing a numeric keypad and some text. The main area of the monitor displays lines of code in a dark-themed editor, with some text highlighted in green. At the bottom of the screen, a taskbar with various application icons is visible.

Introduction To Programming Using C++ Level 2

Bitwise OR Operator



Introduction To Programming Using C++

Level 2

Bitwise OR Operator

Definition

- ◆ The Bitwise OR Operator `|` compares two numbers bit by bit.
- ◆ It returns `1` when at least one of the corresponding bits is `1`.
- ◆ It returns `0` only when both corresponding bits are `0`.

Explanation

-  Computers represent integer values using Binary numbers made of `0` and `1`.
-  The Bitwise OR Operator compares the corresponding bits of two numbers.
-  Each bit is processed independently.
-  If either input bit is `1`, the result for that position is `1`.

Procedures

1. Convert the numbers to Binary.
2. Write the Binary values one below the other.
3. Compare the corresponding bits from left to right.
4. If at least one bit is `1`, write `1`.
5. If both bits are `0`, write `0`.
6. Convert the resulting Binary number back to Decimal if needed.

💡 Example

- ◆ Suppose we have:

```
int Result = 25 | 12;  
cout << Result;
```

- ◆ In Binary:

25 = 00011001

12 = 00001100

00011101

- ◆ The corresponding bits are compared:

00011001

| 00001100

00011101

- ◆ `00011101` in Decimal is `29`.

➡ Result

The result is:

29

Note

- ◆ The Bitwise OR Operator works directly on the individual bits of integer values.
- ◆ The operator used for Bitwise OR is:

`|`

⊘ Warning

- ◆ Bitwise OR `|` is different from Logical OR `||`.
- ◆ `|` works on individual bits, while `||` is used to evaluate logical conditions.



Comparison

▨ Bitwise OR `|`

- ✓ Compares individual bits.
- ✓ Returns `1` when at least one corresponding bit is `1`.
- ✓ Commonly used for bit manipulation and bit masks.

▤ Logical OR `||`

- ✓ Works with logical conditions.
- ✓ Evaluates whether at least one condition is True.
- ✓ Commonly used in `if` statements and other logical expressions.

✦✦ Summary

- ✓ Bitwise OR ``|`` compares numbers bit by bit.
- ✓ A bit becomes ``1`` when at least one corresponding bit is ``1``.
- ✓ ``25 | 12`` produces ``29``.

Thank you 😊